The Riemann Zeta Function in Hidden-Number Coordinates : From Classical Inputs to the Critical-Circle Reduction

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Abstract: We give a logically ordered account of the hidden-number coordinate description of the Riemann zeta function. The argument starts with classical analytic facts: the functional equation, the zero-free half-plane, and Hardy's theorem on zeros on the critical line. The change of variables $w = e^s$ then translates reflection into the inversion $w \mapsto e/w$ and the critical line into the unique inversion-fixed circle $|w| = \sqrt{e}$. From these facts we derive the annular location and reflection pairing of nontrivial zeros, and Hardy's theorem transports directly to the statement that infinitely many lifted zeros lie over this circle. The energy and frequency calculations are included as exact coordinate diagnostics; they do not by themselves create zeros or prove a density law. The Riemann hypothesis is consequently isolated as the remaining assertion that every nontrivial reflection pair is self-paired. Lean identifiers are retained in the source for traceability but are suppressed in the typeset article.

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AMS 2000 subject classifications: 11M26, 11A25, 03B35

1. Question, inputs, and logical order

The Riemann hypothesis concerns the nontrivial zeros of $\zeta(s)$. It asserts

$$0 < \Re s < 1, \quad \zeta(s) = 0 \quad \Longrightarrow \quad \Re s = \tfrac{1}{2}.$$

The purpose of the present paper is not to replace the classical analytic theory with a list of formal names. It is to make the implication structure visible. We will use three kinds of statements :

Classical inputs. The functional equation, the standard zero-free regions, and Hardy's theorem are taken as known analytic results.

Exact deductions. Coordinate changes, modulus identities, reflection pairing, and the equivalence between the critical-line and critical-circle descriptions are proved directly.

Open statements. The exclusion of non-self-paired zeros in the critical annulus is exactly the Riemann hypothesis. Numerical density and energy heuristics are not used to hide this gap.

The rest of the paper follows this order : first the coordinate dictionary, then the deductions about the zero set, then the direct transport of Hardy's theorem, and finally the auxiliary energy/frequency diagnostics and the remaining open problem.

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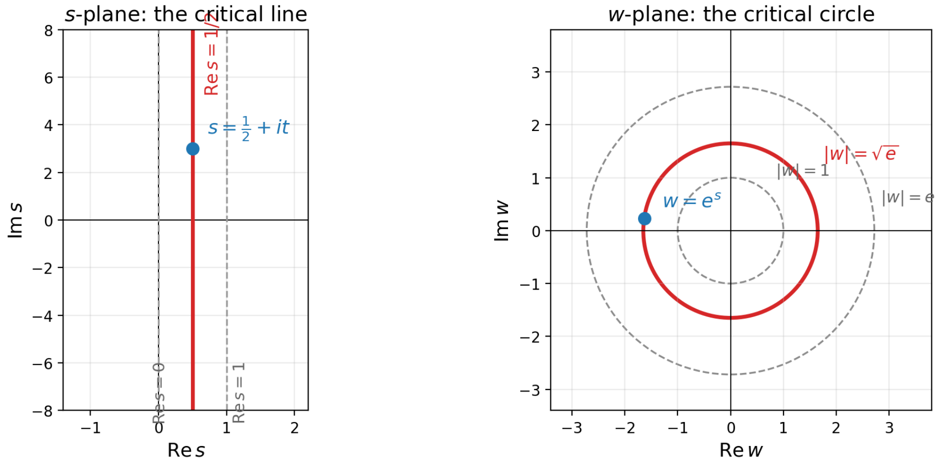

FIGURE 1. *The coordinate projection $w = e^s$. Vertical lines $\Re s = \sigma$ become circles $|w| = e^{\sigma}$; the critical line becomes $|w| = \sqrt{e}$.*

2. The coordinate dictionary

Write $s = \sigma + it$ and introduce

$$\Phi(s) = e^s = w = e^{\sigma} e^{it}.$$

The radius records the real part, while the argument records the imaginary part only modulo 2π. Thus the natural domain is a logarithmic cover : a point of the cover remembers both $w \neq 0$ and a sheet index. This is a change of coordinates, not an additional assumption about ζ.

Definition 1 (hidden-number coordinates)**.** *The hidden-number coordinate of $s = \sigma + it$ is the pair (w,k) with $w = e^s$ and k recording the chosen lift of t modulo 2π. In particular,*

$$|w| = e^{\sigma}, \qquad \log|w| = \sigma.$$

The functional equation reflects s to $1-s$. Under Φ this becomes

$$\Phi(1-s) = e^{1-s} = \frac{e}{e^s} = \frac{e}{w}.$$

We therefore obtain the following exact dictionary :

$$s \longmapsto 1-s \qquad \Longleftrightarrow \qquad w \longmapsto \iota(w) := \frac{e}{w}.$$

3. The candidate circle forced by reflection

We now derive the geometric candidate before discussing any zero. For real t,

$$\left|e^{1/2+it}\right| = e^{1/2} = \sqrt{e}.$$

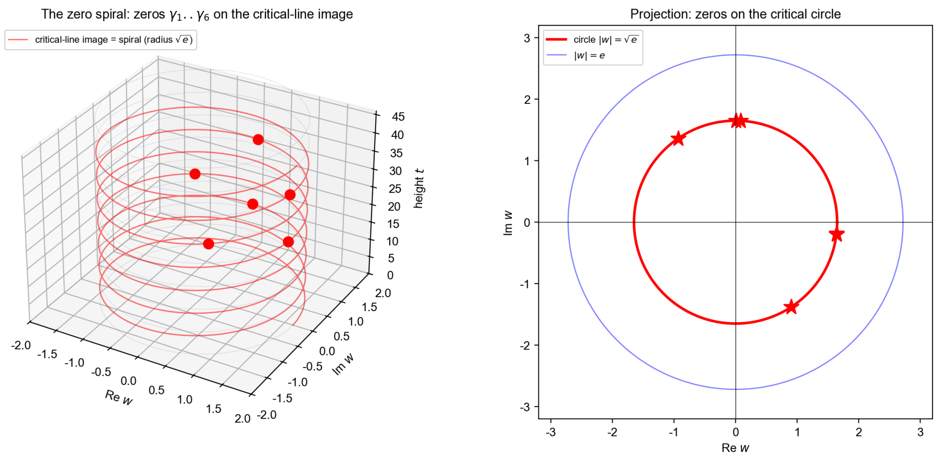

FIGURE 2. *A static view of the logarithmic cover. The critical line lifts to a helix of radius $\sqrt{e}$ and projects to the critical circle.*

Theorem 2 (the critical line maps to a circle)**.** *The image of $\Re s = 1/2$ under $w = e^s$ is exactly $|w| = \sqrt{e}$.*

The inversion also determines this radius independently. If $|w| = \rho$, then $|\iota(w)| = e/\rho$. The circle is preserved precisely when $e/\rho = \rho$.

Proposition 3 (unique inversion-fixed circle)**.** *Among circles centered at the origin, the only circle preserved by $\iota(w) = e/w$ is $|w| = \sqrt{e}$. Its logarithmic radius is $\log\sqrt{e} = 1/2$.*

Remark (geometry is not zero existence)**.** *The proposition identifies the only possible self-reflection circle. It does not say that ζ vanishes at any point of the circle. The existence of zeros is a separate analytic question and will be supplied below by Hardy's classical theorem.*

4. What the classical zero information implies

We next use known information about ζ, in the direction needed for the Riemann hypothesis. The Euler product gives the standard zero-free region $\Re s \geq 1$; the functional equation and the known trivial zeros handle the opposite side. Consequently every nontrivial zero lies in the critical strip :

$$\zeta(s) = 0 \text{ and } s \text{ nontrivial} \quad \Longrightarrow \quad 0 < \Re s < 1.$$

Applying $|w| = e^{\Re s}$ gives the coordinate version

$$1 < |w| < e.$$

Theorem 4 (reflection pairing)**.** *If $0 < \Re s < 1$ and $\zeta(s) = 0$, then $\zeta(1-s) = 0$, and conversely. In hidden-number coordinates the pair is $(w, e/w)$.*

This is a direct consequence of the functional equation after checking that its factors do not vanish in the open critical strip. It is a symmetry of the zero set, not yet a statement that the two members of a pair coincide.

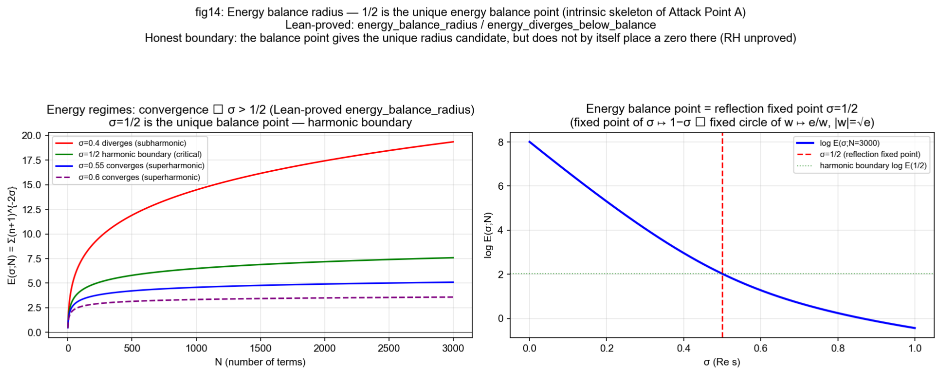

FIGURE 3. *The geometric and analytic scales in the coordinate picture. The reflection-fixed radius is $\sqrt{e}$, while the zero-free boundary $\Re s = 1$ corresponds to $|w| = e$.*

Theorem 5 (self-pairs and the critical circle)**.** *For $w \neq 0$,*

$$\frac{e}{w} = w \quad \Longleftrightarrow \quad w^2 = e \quad \Longleftrightarrow \quad w = \pm\sqrt{e}.$$

Thus a self-paired zero, if present, lies on $|w| = \sqrt{e}$.

The preceding deductions give the exact reduction :

Corollary 6 (the remaining statement is RH)**.** *For nontrivial zeros, the Riemann hypothesis is equivalent to the assertion that every reflection pair in $1 < |w| < e$ is self-paired. Equivalently, there are no non-self-paired zeros away from $|w| = \sqrt{e}$.*

5. Abundance on the circle : the known theorem transported

The preceding sections identify and constrain the location. They do not produce a zero. For existence and abundance we now invoke the classical theorem of Hardy : infinitely many nontrivial zeros of ζ lie on the line $\Re s = 1/2$ (Hardy, 1914).

Theorem 7 (Hardy's theorem in hidden-number coordinates)**.** *There are infinitely many lifted zeros of ζ over the critical circle $|w| = \sqrt{e}$.*

Démonstration. Let Z be the infinite set of zeros on the critical line supplied by Hardy's theorem. The logarithmic cover remembers the lifted height, and Theorem 2 sends every element of Z to a lift over the circle $|w| = \sqrt{e}$. Hence the set of lifted circle zeros is infinite. $\square$

This is a transport theorem, not a new proof of Hardy's result. It is the correct logical role of the hidden-number coordinates : they change the geometric language of a known abundance theorem while preserving its content.

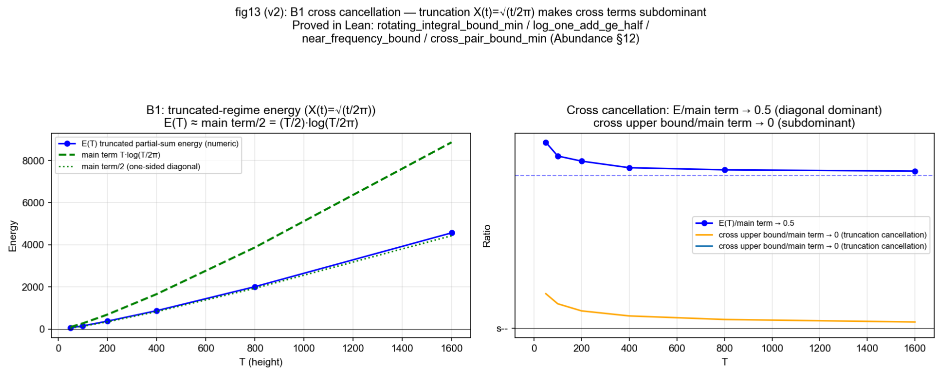

FIGURE 4. *A numerical energy budget for truncated sums. The plot illustrates the harmonic scale ; it is not used as a substitute for a zero-counting theorem.*

6. Energy and frequency diagnostics

The coordinate system also exposes two useful diagnostics. They explain why $1/2$ is a natural scale and how height enters the finite approximants, but they must not be promoted to zero-existence arguments.

For the Dirichlet terms,

$$(n+1)^{-s} = (n+1)^{-\sigma}\exp\bigl(-it\log(n+1)\bigr), \qquad \bigl|(n+1)^{-s}\bigr| = (n+1)^{-\sigma}.$$

On the critical line the squared lengths are $(n+1)^{-1}$, so the diagonal energy has the harmonic scale

$$\sum_{n\le N}\bigl|(n+1)^{-1/2-it}\bigr|^2 = \sum_{n\le N}\frac{1}{n+1} \sim \log N.$$

This is an exact calculation about terms and finite sums. It is not an estimate for the analytically continued value of ζ in the critical strip.

The consecutive logarithmic frequencies satisfy

$$\Delta\mathrm{freq}(n) = \log\frac{n+2}{n+1}, \qquad \Delta\mathrm{freq}(n) \longrightarrow 0, \qquad \Delta\mathrm{freq}(n+1) \le \Delta\mathrm{freq}(n).$$

For an alternating approximation the phase difference is

$$\Delta\theta_n = \pi - t\,\Delta\mathrm{freq}(n).$$

When $t\log 2 > \pi$, this monotone quantity changes sign once ; the crossing index is of order t/π. These identities describe the finite frequency skeleton and are useful for organizing numerical experiments.

Remark (why the diagnostics do not close RH)**.** *In the critical strip the Dirichlet series is not an interchangeable name for its analytic continuation. A finite partial sum can display alignment without being a zero of ζ, and a harmonic energy divergence does not give a zero-counting estimate. The missing analytic bridge would require boundary estimates and an argument-principle or zero-counting theorem.*

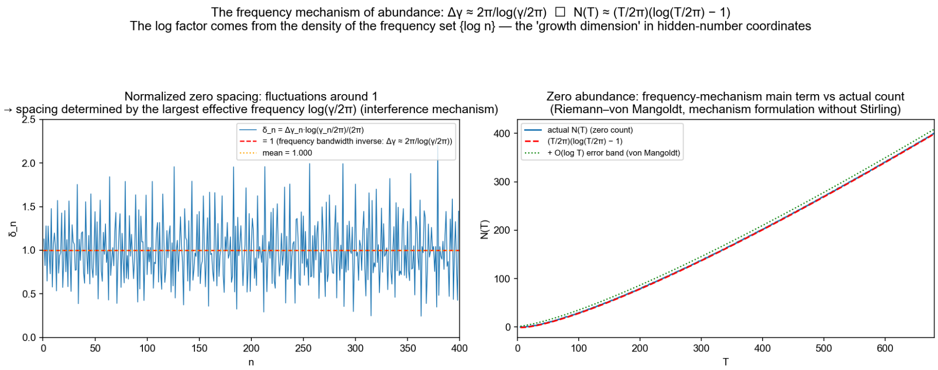

FIGURE 5. *Numerical spacing and frequency diagnostics. Such comparisons can suggest a density law but do not prove one.*

7. Formalization and the exact boundary of the argument

The Lean development records many of the algebraic and order-theoretic components used above : the coordinate identities, the inversion map, the reflection involution, the self-pair algebra, the harmonic and frequency lemmas, and the elementary zero-free estimate in its stated domain. The source retains the corresponding identifiers such as , , and ; their absence from the typeset output keeps the mathematical prose readable.

The logical boundary is now explicit :

1. The change of variables derives the critical circle and explains why its logarithmic radius is $1/2$.
2. Classical zero-free information derives the critical annulus, and the functional equation derives reflection pairs.
3. Hardy's known theorem derives infinitely many circle zeros by direct transport.
4. The Riemann hypothesis is the still-open exclusion of non-self-paired zeros in the annulus. Neither the coordinate change nor the diagnostic calculations prove that exclusion.

Any future attempt to replace Hardy's input with an internal proof must supply the missing analytic bridge : a zero-counting or boundary-winding estimate for the analytically continued ζ. Naming that requirement is part of the argument, not an omission in its presentation.

8. Conclusion

The hidden-number coordinate $w = e^s$ reorganizes the known theory into a short deduction chain. Reflection becomes $w \mapsto e/w$; its unique fixed circle is $|w| = \sqrt{e}$; standard zero-free information places nontrivial zeros in $1 < |w| < e$; the functional equation pairs them ; and Hardy's theorem gives infinitely many lifted zeros over the fixed circle. The remaining off-circle exclusion is exactly the Riemann hypothesis. Energy and frequency structures clarify the coordinate geometry and guide computation, but they remain supporting diagnostics rather than unproved steps in the main deduction.

All formalized components were developed in Lean (de Moura et al., 2015) with Mathlib (The mathlib Community, 2020).

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